



S.P.M College, Udantpuri

Bachelor Of Computer Application (BCA)

Part -2 (Paper-4)

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IP Addressing



IP Address & MAC Address

❖ What is MAC address ?

- MAC stands for Media Access Control.
- A MAC address is a unique address used to identify a network interface/device on a local network.
- A standard MAC address is 48 bits (6 bytes) long. 1 byte = 8 bits, so:
 $6 \text{ bytes} \times 8 = 48 \text{ bits}$
- A standard MAC address contains 12 hexadecimal digits (0–9 and A–F). Example = 00:1A:2B:3C:4D:5E
- Each hexadecimal digit represents 4 bits. Therefore : $12 \text{ hexadecimal digits} \times 4 = 48 \text{ bits}$
- MAC address is also commonly called: Physical Address / Hardware Address / Ethernet Address
- MAC address is associated/linked with a Network Interface Card (NIC) or network interface.
- MAC addressing is associated / linked with the Data Link Layer (Layer 2) of the OSI Model.

- The MAC address is used for communication within a local network (LAN).

MAC

Media Access Control Address



Organizationally Unique Identifier Network Interface Controller Specific

- A MAC address is generally written in one of these formats:
00:1A:2B:3C:4D:5E { use colon notation, See - linux/Unix/MAC OS)
00-1A-2B-3C-4D-5E { use hyphen notation, See - Windows }
001A.2B3C.4D5E {period-separated notation, See – Cisco / Networking devices)
- The first 24 bits (6 hexadecimal digits) are called the OUI (Organizationally Unique Identifier). The OUI identifies the organization/manufacturer that was assigned that address block.
- The remaining 24 bits (6 hexadecimal digits) identify the particular network interface /device identifier within that assigned block.
- MAC addresses are assigned under standards administered by the IEEE 802 family.

Note : However, a MAC address is not absolutely permanent. Modern operating systems / Software can use MAC randomization or allow a MAC address to be changed (spoofed).

- A device can have more than one MAC address if it has multiple network interfaces.

For example, a computer may have:

Ethernet/LAN interface → one MAC address

Wi-Fi interface → another MAC address

Bluetooth interface → another address

- IP address and MAC address are different.
- MAC addresses are used by switches/bridge to identify

devices/interfaces on a LAN.

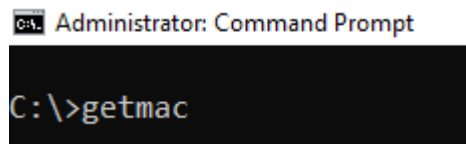
- A switch maintains a MAC Address Table (MAC/CAM table).
- The switch learns which MAC address is available on which port by examining the source MAC address of incoming frames.
- When forwarding an Ethernet frame, the switch uses the destination MAC address to decide where to send the frame.
- ARP (Address Resolution Protocol) is used in IPv4 networks to find the MAC address associated with an IP address on the local network.
- ARP is therefore a mechanism that helps map:
IPv4 Address → MAC Address

Example:

192.168.1.10 → ARP → 00:1A:2B:3C:4D:5E

To find the MAC address in Windows, you can use:

getmac

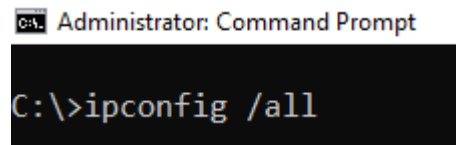


Administrator: Command Prompt

C:\>getmac

You can also use:

ipconfig /all



Administrator: Command Prompt

C:\>ipconfig /all

- In ipconfig /all, the MAC address is usually shown as Physical Address.
- You can also check the MAC address through the Network Adapter/NIC settings of the operating system.
- The MAC address belongs to the network interface, not necessarily to the entire computer as a single identity.

To find the MAC address in mobile, you can use

Settings → About Phone → Status / Status Information → Wi-Fi MAC Address & Bluetooth Address (or)

Settings → About Phone → Detailed Info and Specs → Status → Wi-Fi MAC Address (or) Settings → Network & Internet → Wi-Fi → Connected Wi-Fi → MAC Address

But use Mobile data then not provide MAC address. It provide Cellular interface (IMEI) and your device identifiers from cellular network to help SIM.

History & Founder

- MAC addressing originated with Ethernet development at **Xerox PARC in the 1970s** and was later standardized by IEEE, becoming essential for identifying network interfaces at the Data Link Layer.

C:\Users\hira>ipconfig /all

Windows IP Configuration

Host Name : hira
Primary Dns Suffix :
Node Type : Hybrid
IP Routing Enabled. : No
WINS Proxy Enabled. : No

Ethernet adapter Ethernet:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Realtek PCIe GbE Family Controller
Physical Address. : 48-BA-4E-A -EF-23
DHCP Enabled. : No
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 1:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter
Physical Address. : D6-6A-6A-1 -46-0F
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Local Area Connection* 10:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Microsoft Wi-Fi Direct Virtual Adapter #2
Physical Address. : D4-6A-6A-1 -46-0F
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . :
Description : Realtek RTL8723DE 802.11b/g/n PCIe Adapter
Physical Address. : D4-6A-6A-1 -46-0F
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes
IPv6 Address. : 2401:4900:8909:9577:7282:6853:ada9:34d8(Preferred)
Temporary IPv6 Address. : 2401:4900:8909:9577:950c:da51:c502:a6ca(Preferred)
Link-local IPv6 Address : fe80::7c86:daa8:6c78:d85f%4(Preferred)
IPv4 Address. : 192.168.1.8(Preferred)
Subnet Mask : 255.255.255.0
Lease Obtained. : Wednesday, September 9, 2026 12:11:45 AM
Lease Expires : Thursday, September 10, 2026 6:42:52 AM
Default Gateway : fe80::6e4f:89ff:fee4:8e29%4
192.168.1.1
DHCP Server : 192.168.1.1
DHCPv6 IAID : 64252522
DHCPv6 Client DUID. : 00-01-00-01-2F-16-5E-83-48-BA-4E-AB-EF-23
DNS Servers : fe80::6e4f:89ff:fee4:8e29%4
192.168.1.1
fe80::6e4f:89ff:fee4:8e29%4
Primary WINS Server : 192.168.1.1
NetBIOS over Tcpip. : Enabled

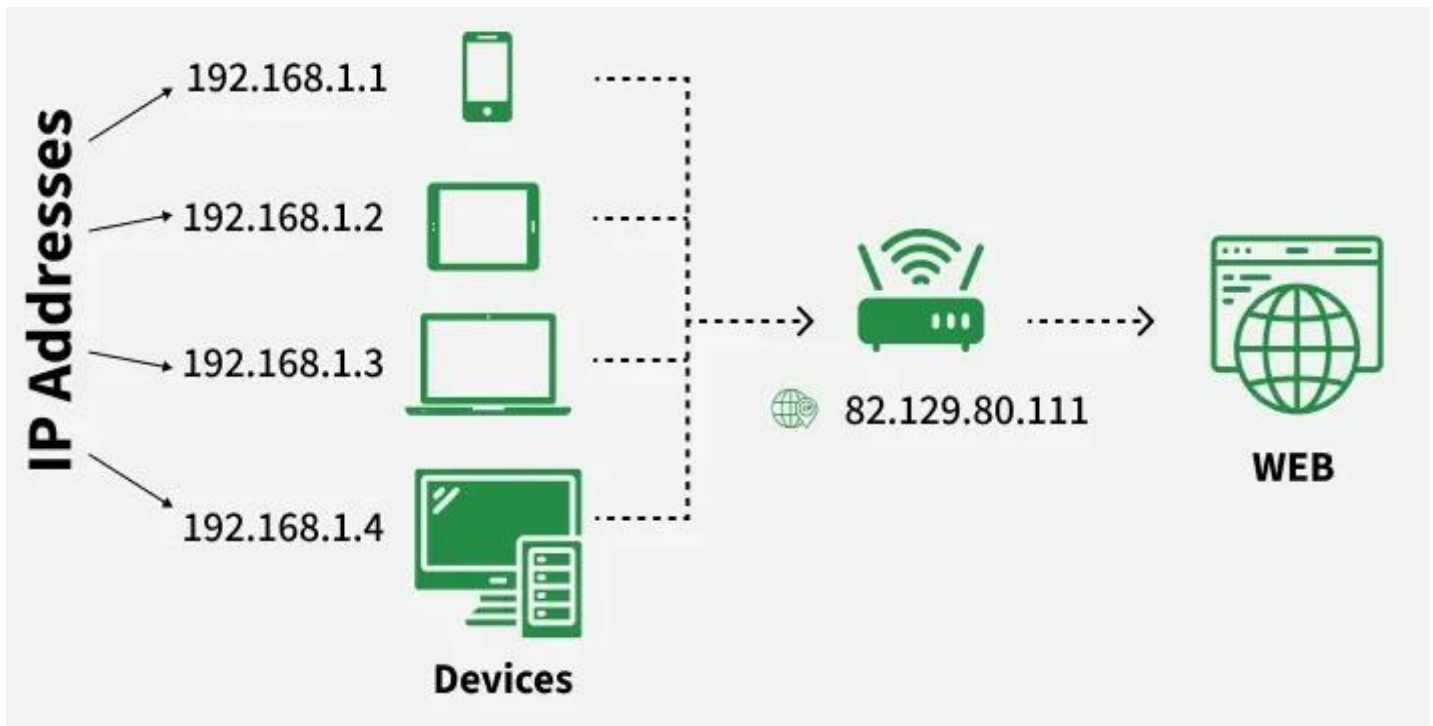
Ethernet adapter Bluetooth Network Connection:

Media State : Media disconnected
Connection-specific DNS Suffix . :
Description : Bluetooth Device (Personal Area Network)
Physical Address. : D4-6A-6A-1E-45-10
DHCP Enabled. : Yes
Autoconfiguration Enabled : Yes

C:\Users\hira>

❖ What is IP Addressing ?

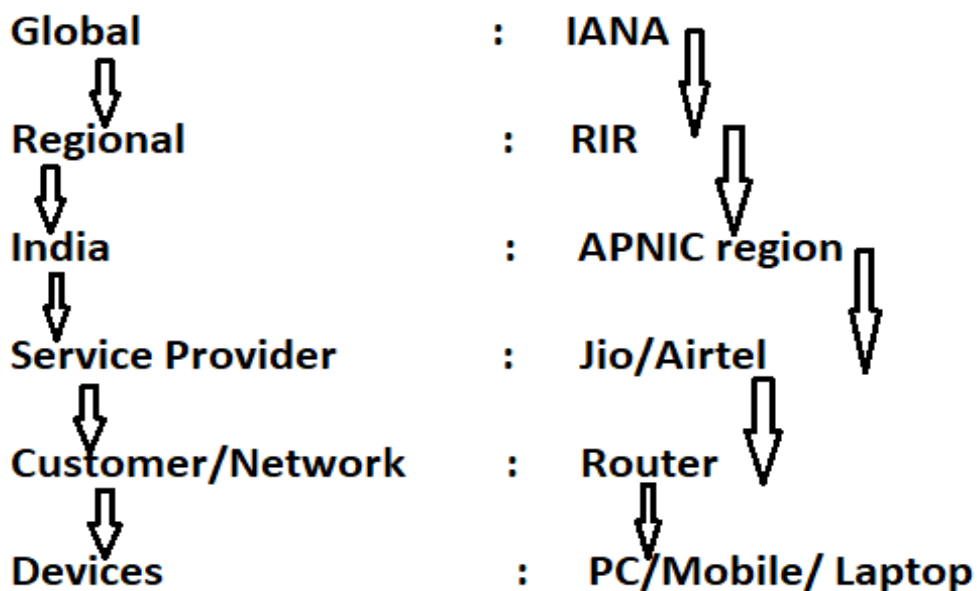
- IP means Internet protocol. IP addressing is a method of assigning a unique logical address (IP address) to each device on a network so that devices can identify and communicate with each other.
- IP addressing is mainly studied under the Network Layer (Layer 3) of the OSI model.



❖ History of IP address

- ✓ **1977** : Vint cerf & Robert kahn designed TCP/IP, and Vint cerf chose 32-bit IPv4. SO, Founder of IP is Vint cerf. Developed by – DARPA (Defense Advanced Research Projects Agency)
- **1981** :IPv4 become official standard (RFC 791).
- **1990s** : Address shortage led to CDIR & NAT.
- **2011** : Final IPv4 blocks allocated.
- **2012** : IPv6 (128 bit) adopted all companys and solve shortage permanently.IPv6 work began in 1998.

❖ Control & manage IP address



➤ **IANA (Internet Assigned Numbers Authority)**

- IANA coordinates global IP address space and other Internet number resources.
- IANA does not normally assign an IP directly to your phone or computer. It allocates large address blocks to regional organizations called RIRs.

➤ **RIR (Regional level)**

- RIR receive address resources (IANA) and distribute them within their regions
- In World total 5 major regional internet registries.
 - (a) **APNIC (Asia- pacific)**
 - (b) **RIPE NCC (Europe, Middle east, central asia)**
 - (c) **ARIN (USA, Canada and parts of the Caribbean)**
 - (d) **LACNIC(Latin America and caribbean)**
 - (e) **AFRINIC (Africa)**
- India comes under APNIC.

➤ **ISP (Internet Service provider)**

- ISPs such as Jio, Airtel, BSNL, etc obtain manage IP address resources and use them for their network and customers.
- APNIC → ISP → Your router/mobile network → your device

❖ It includes topics such as:

1. IPv4 and IPv6
2. IP address classes
3. Network ID and Host ID
4. Subnet mask
5. Subnetting
6. Public and Private IP addresses
7. Static and Dynamic IP addresses (DHCP)

❖ Types of IP Based on -

1. Protocol Version

Two types (a) IPv4 (b) IPv6

2. Assignment Method

Two types (a) Static IP (b) Dynamic IP

3. Visibility and Scope

Two types (a) Public IP (b) Private IP

4. Network Classes (IPv4 Classful Addressing)

Five types (a) Class A (b) Class B (c) Class C (d) Class D (e) Class E

5. Operational Characteristics

Four types (a) Unicast (b) Multicast (c) Broadcast (d) anycast

❖ Types of IP address based on protocol version

Two types of ip address

- (i) IPv4 (internet protocol version 4)
- (ii) IPv6 (internet protocol version 6)

1. IPv4 Address

- IPv4 stands for Internet Protocol Version 4.
- An IPv4 address is a 32-bit (4-byte) logical address.
- It is normally written in decimal (dotted-decimal) notation.
Example:- 192.168.1.10

- An IPv4 address is also called a Logical Address.
- IPv4 operates at Layer 3 – Network Layer of the OSI Model.
- IPv4 is used to identify network interfaces and enable communication between different networks.
- An IP address can be dynamic or static. A dynamic IP may change over time.
- Total ip available in IPv4 = 4.3 billion addresses
32 bits $\rightarrow 2^{32} \rightarrow 4.3$ billion address (approx.)

➤ **To check the local IP address in Windows, use:**

c:\> Ipconfig

C:\> Administrator: Command Prompt

C:\>ipconfig

Wireless LAN adapter Wi-Fi:

Connection-specific DNS Suffix . . :

IPv6 Address. : 2401:4900:8909:9577:7282:6853:ada9:34d8

Temporary IPv6 Address. : 2401:4900:8909:9577:950c:da51:c502:a6ca

Link-local IPv6 Address : fe80::7c86:daa8:6c78:d85f%4

IPv4 Address. : 192.168.1.8

Subnet Mask : 255.255.255.0

Default Gateway : fe80::6e4f:89ff:fee4:8e29%4
192.168.1.1

OR

To check your public IP address, you can use a search engine with:

“What is my IP address?”

Google what is my ip address

whatismyip.unc.edu

What Is My IP?

Network diagnostics · UNC Information Technology Services

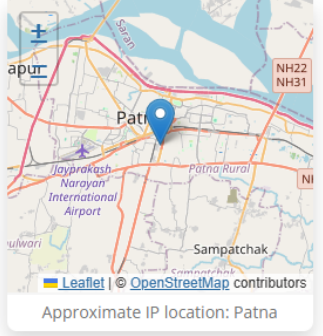
HOME SPEED TEST CONNECTIVITY FAQ ABOUT

You are connected from off campus over the Internet.

PRINT REPORT (PDF)

IPv6 2401:4900:8909:9577:950c:da51:c502:a6ca

IPv4 223.235.103.146



Approximate IP location: Patna

- ARP (Address Resolution Protocol) is used in IPv4 networks to find the MAC address corresponding to an IPv4 address on the local network.

IP to MAC mapping → ARP

- RARP (Reverse Address Resolution Protocol) performs the reverse mapping conceptually:

MAC to IP mapping → RARP

Note : RARP is largely obsolete and has been replaced by mechanisms such as DHCP.

2. IPv6 Address

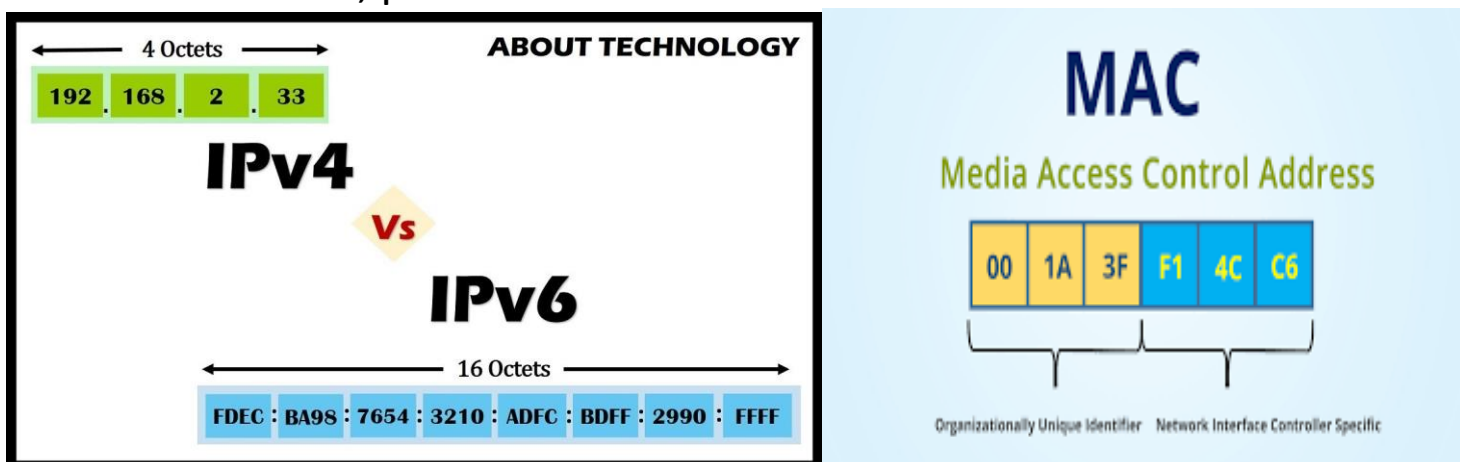
- 128 bits / 16 bytes
- IPv6 Address is also called a Logical Address.
- IPv6 uses Hexadecimal notation.
Example : 2401:4900:8909:9577:950c:da51:c502:a6ca
- IPv6 operates at Layer 3 – Network Layer of the OSI Model.
- IPv6 was developed and standardized by the IETF (Internet Engineering Task Force).
- IPv6 development work began in the 1990s, and its specification was standardized in 1998.

- IPv6 addresses can change depending on network configuration.
- Use: To identify network interfaces and enable communication over IPv6 networks.
- To check IPv6 on Windows, use ipconfig; public IPv6 can also be checked using an Internet search.
- IPv6 uses NDP (Neighbor Discovery Protocol) instead of ARP/RARP for neighbor discovery and link-layer address resolution

❖ Change IP Address

Different ways to change an IP address

1. **Switch Networks / Contact Your ISP** – Connecting to another network may give you a different IP address. Your ISP can also change or reassign your public IP.
2. **Restart/Reset Your Modem or Router** – With a dynamic public IP, reconnecting may result in a different IP, depending on your ISP.
3. **Connect via a VPN** – A VPN routes your Internet traffic through its server, so websites generally see the **VPN server's public IP address** instead of your own.
4. **Use a Proxy Server** – A proxy acts as an **intermediary server** between your device and the destination; websites may see the proxy's IP instead of your device's public IP.
5. **Change the Local IP Manually** – On a LAN, you can configure a different private IP address, such as changing 192.168.1.10 to 192.168.1.20, provided the new address is valid and available.



❖ Types of IP address based on assignment/ configuration

Two types of IP address

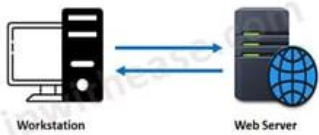
1. Static IP Address

- A Static IP address is an IP address that is manually assigned or reserved and normally does not change.
- Example: 192.168.1.10
- Used for: Servers, printers, CCTV cameras, routers, and other devices that need a fixed address.

2. Dynamic IP Address

- A Dynamic IP address is an IP address that is automatically assigned by a DHCP server and may change over time.
- Example:
Today → 192.168.1.10
Later → 192.168.1.15
- Used for: Mobile phones, laptops, home computers, and most ordinary network devices.

Short Difference

Function	Static IP address	Dynamic IP address
		
Provider	Internet Service Provider	DHCP (Dynamic Host Configuration Protocol)
Nature	Constant	Not constant
Security	Less secure	More secure
Traceability	Easily traceable	Difficult to trace
Stability	Stability is higher	Less stable
Costs	Costly	No direct cost Cost associated to setup initial infrastructure DHCP server
Confidentiality	Not so confidential	Higher security
Troubleshooting	Easy	Complex

❖ Types of IPv4 Classes

- **IP address classes** are the traditional IPv4 classification system used to divide IPv4 addresses into different network sizes.
- This is divided into 5 (Five) classes

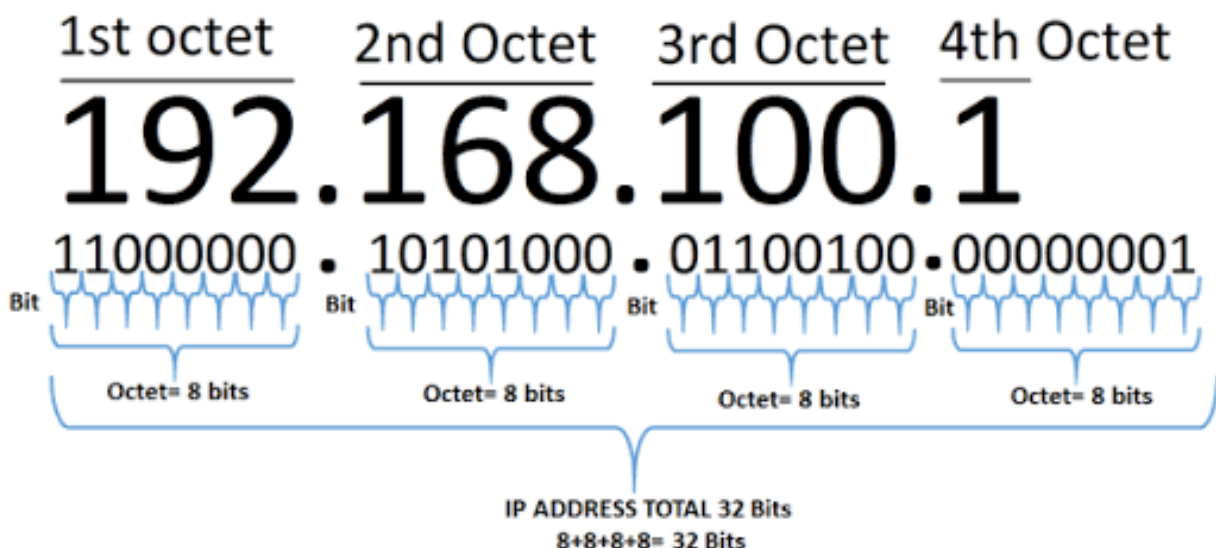
1. Class A
2. Class B
3. Class C
4. Class D
5. Class E

❖ Why were classes needed?

IP classes were created to divide IPv4 address space into different network sizes according to the number of networks and hosts required.

❖ What is Octet and Octet Range?

- Each part/byte of an IPv4 address is called an octet, and each octet can contain a specific range of values, known as the octet range.
- An IPv4 address consists of 4 octets, separated by dots (.).
- Each octet contains 8 bits (1 byte).
- Minimum value: 00000000 = 0, Maximum value: 11111111 = 255
- The range of one octet is 0 to 255.
- The maximum value of one octet is calculated as: $2^8 - 1 = 256 - 1 = 255$. i.e. **1 Octet = 8 bits = $2^8 = 256$ possible values (0–255)**
- Therefore, each octet can contain a value from 0 to 255.
- The complete IPv4 address range is: 0.0.0.0 to 255.255.255.255
- If any octet contains a value greater than 255, the IPv4 address is invalid.
- Example of a valid address: 192.168.1.100 (Right)
- Example of an invalid address: 192.168.1.300 (Wrong)



❖ IPv4 Address Classes Range (Decimal):

Class	First Octet Range	All IP range	Use
Class A	1-126	1.0.0.0 – 126.255.255.255	Big Network
Class B	128-191	128.0.0.0-191.255.255.255	Middle Net
Class C	192-223	192.0.0.0-223.255.255.255	Small Net
Class D	224-239	224.0.0.0-239.255.255.255	Multicast
Class E	240-255	240.0.0.0-255.255.255.255	Reserved/ Experimental

❖ Identify IP class range In Binary represent

1 st Octet	Class
0	Class A
10	Class B
110	Class C
1110	Class D
1111	Class E

Types of IP Addresses (Base Representations)					
Class	IPv4 (Binary)	IPv4 (Dotted Decimal)	IPv4 (Hexadecimal)	IPv6 (Compressed)	Binary Range Pattern (IPv4)
Class A	01111111 00000001	127.1.1.1	0x7F010101	::	0xxxxxxx ...
Class B	10111111 00000001	191.1.1.1	0xBF010101	::	10xxxxxx ...
Class C	11011111 00000001	223.1.1.1	0xDF010101	::	110xxxxx ...
Class D	11100000 ...	224.0.0.1	0xE0000001	::	1110xxxx ...
Class E	11110000 ...	240.0.0.1	0xF0000001	::	1111xxxx ...
Class A	IPv6 Subnets				
Class B	01111111 ...	127.1.1.1	0x7F010101	::	
Class C	10111111 ...	223.1.1.1	0xBF010101	::	
Class D	11011111 ...	224.0.0.1	0xE0000001	::	
Class E	11110000 ...	240.0.0.1	0xF0000001	:::1	

❖ What is Loopback / Loopback address?

- Loopback is a networking technique in which data packets or datagrams are sent back to the same device without leaving the device or going onto the external network.
- It is mainly used for testing and troubleshooting network applications and services.
- **Loopback allows a device to send data to and receive data from itself using a loopback address.**

- For example, when we develop a website or web application, we can test it locally before making it public.
- The most commonly used IPv4 loopback address is 127.0.0.1, which is also called localhost.
- IPv4 Loopback Range: 127.0.0.0 – 127.255.255.255
- The entire 127.0.0.0/8 block is reserved for loopback purposes.

➤ **Key points**

- ✓ If booting (start/restart) IP address → 0.0.0.0
- ✓ Loop back address/ reserved / local host → 127.0.0.1 – 127.255.255.255
- ✓ Broadcast IP address – 255.255.255.255
- ✓ Convert domain name (google.com / amazon.com/ppup.ac.in) into IP address → DNS (domain name system)
- ✓ DNS generate IP address
- ✓ Automatic IP address to host in network – DHCP (Dynamic host configuration protocol)

❖ **IPv4 Address Classification — 40 Questions (Decimal based)**

S.No.	IP Address	Class A	Class B	Class C	Class D	Class E	Invalid
1	10.25.36.7						
2	198.45.12.8						
3	172.16.5.4						
4	224.1.2.3						
5	25.100.20.5						
6	255.255.256.255						
7	150.23.45.1						
8	240.5.6.7						
9	8.8.8.8						
10	203.01.113.5						
11	180.50.20.10						
12	127.0.0.1						

S.No.	IP Address	Class A	Class B	Class C	Class D	Class E	Invalid
13	192.168.100.25						
14	230.10.20.30						
15	260.2.32.1						
16	100.20.30.40						
17	19.0.0.1						
18	239.255.255.255						
19	128.0.0.1						
20	192.168.1						
21	11.22.33.44						
22	191.255.255.254						
23	225.10.10.10						
24	250.1.2.3						
25	172.31.255.255						
26	300.10.20.30						
27	210.20.30.40						
28	126.255.255.254						
29	0.255.255.255						
30	145.20.30.40						
31	255.255.255.255						
32	200.10.20.30						
33	127.255.255.255						
34	223.255.255.254						
35	240.10.20.30						
36	192.168.500.10						
37	16.20.30.40.50						
38	239.1.1.1						
39	130.40.50.60						
40	10.10.10.255						

Answer

S.No. Answer

1 Class A
 2 Class C
 3 Class B
 4 Class D
 5 Class A
 6 Invalid
 7 Class B
 8 Class E
 9 Class A
 10 Class C
 11 Class B
 12 Reserved/Loopback
 13 Class C
 14 Class D
 15 Invalid
 16 Class A
 17 Class A
 18 Class D
 19 Class B
 20 Invalid

S.No. Answer

21 Class A
 22 Class B
 23 Class D
 24 Class E
 25 Class B
 26 Invalid
 27 Class C
 28 Class A
 29 Reserved/Special
 30 Class B
 31 Reserved/Broadcast
 32 Class C
 33 Reserved/Loopback
 34 Class C
 35 Class E
 36 Invalid
 37 Invalid
 38 Class D
 39 Class B
 40 Class A

❖ **IPv4 Address Classification — 20 Questions
 (Binary based)**

No. Binary IPv4 Address**Class**

1 00001010.00010100.00011110.00101000 ?
 2 01000001.00001010.00010100.00000101 ?
 3 01111111.00001010.00001010.00000001 ?
 4 10000000.00001010.00010100.00000001 ?
 5 10010110.00011001.00001010.00000101 ?
 6 10111111.00001010.00010100.00011110 ?

No. Binary IPv4 Address**Class**

- 7 11000000.10101000.00000001.00000001 ?
8 11001000.00001010.00010100.00000101 ?
9 11011111.00001010.00001010.00001010 ?
10 11100000.00001010.00010100.00000001 ?
11 11100110.00001010.00010100.00000001 ?
12 11101111.11111111.00000001.00000001 ?
13 11110000.00001010.00010100.00000001 ?
14 11110101.00001010.00010100.00000001 ?
15 11111111.00001010.00001010.00001010 ?
16 00000001.00010100.00011110.00101000 ?
17 10000010.00010100.00001010.00000101 ?
18 11000001.00001010.00010100.00011110 ?
19 11100011.00001010.00010100.00000001 ?
20 11111010.00000001.00000001.00000001 ?

Answer Key	
1. A	11. D
2. A	12. D
3. Invalid / Reserved	13. E
4. B	14. E
5. B	15. Invalid / Reserved
6. B	16. A
7. C	17. B
8. C	18. C
9. C	19. D
10. D	20. E

=====HIRA KUMAR=====